ASJID SIDDIQUE

Software Engineering Undergraduate | Machine Learning & Computer Vision | Trustworthy AI & LLM Applications

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SUMMARY

Fifth-semester BS Software Engineering undergraduate at **NUST** (**CGPA 3.76/4.00**) with a foundation in algorithms, statistics, and linear algebra. Builds and evaluates **machine learning** and **computer vision** systems, including **PCBDefect-X** (PCB defect detection) and **FraudShield** (explainable fraud detection), with **reproducible pipelines**, **leakage-aware held-out evaluation**, and deployment. Studied **LLM applications** (**LangChain**, **RAG**, agents) and interested in **trustworthy AI for software engineering**. Seeks to learn research methods, especially experiment design and error analysis, under expert supervision.

RESEARCH INTERESTS

Trustworthy & Reliable AI • AI for Software Engineering • LLM Agents & Tool Use • LLM Evaluation • Machine Learning & Deep Learning • Computer Vision • Reliable Evaluation & Generalization • Explainable AI • Mathematical & Statistical Learning Methods • AI for Scientific Discovery

EDUCATION

National University of Sciences and Technology (NUST), Pakistan *Sep 2024 – Present*
BS Software Engineering | CGPA: 3.76/4.00 | 70 credits completed | Expected June 2028

Punjab College, Burewala *2022 – 2024*
F.Sc. Pre-Engineering | Marks: 1083/1200

SELECTED MACHINE LEARNING & COMPUTER VISION PROJECTS

PCBDefect-X — *Deep Learning & Computer Vision* Demo | Code | Project Write-up *May 2026 – Present*
Python • PyTorch • OpenCV • NumPy • Pandas • RF-DETR • ONNX Runtime • TensorRT • FastAPI • Next.js

- Developed an end-to-end **deep learning** and **computer vision** system for automated PCB defect detection and localization using the **DeepPCB** dataset and **RF-DETR Nano**.
- Built a **reproducible** data pipeline: deterministic dataset conversion, train/validation/test splits **checked for overlapping images** (by path and SHA-256 hash), preprocessing, augmentation, and a fixed random seed.
- Evaluated on a **held-out test set** of 366 images across 6 defect classes: **78.8% mAP@50**, 39.3% mAP@50:95, **90.0% precision**, 74.8% recall, and **81.4% F1-score**, with per-class and localization **error analysis**.
- Exported the model for **ONNX Runtime** and **TensorRT** FP16 inference, **verifying each backend's outputs against the PyTorch model** and running a smoke test.

FraudShield — *Machine Learning & Explainable AI* Demo | Code | Project Write-up *Jun – Aug 2026*
Python • Scikit-learn • PyTorch • SHAP • Pandas • NumPy • FastAPI • Next.js

- Developed an end-to-end machine learning system for highly imbalanced fraud detection on the public creditcard dataset of **284,807 transactions** (about **0.17% fraud**; 283,726 rows after removing 1,081 duplicates).
- Used **train-only preprocessing to avoid data leakage**. Chose the threshold on the validation set with cost-sensitive **threshold optimization** (illustrative cost assumptions), then evaluated **once on an untouched test set**.
- Compared Logistic Regression, Random Forest, HistGradientBoosting, and a PyTorch MLP, trained with SMOTE or class weighting. The selected Random Forest reached **PR-AUC 0.803**, with **86.9% precision, 76.8% recall, and 81.6% F1-score** at the optimized threshold.
- Added **SHAP explanations** of the model's predictions (not causal claims) and a **FastAPI** inference service behind a Next.js interface.

AWARDS & HONORS

Certificate of Achievement — *Punjab College, Burewala* View *2025*

CERTIFICATIONS & ACTIVITIES

AI Basic to Advance Course — *ITSOLERA PVT LTD (Certificate of Course Completion)* View *May – Dec 2025*

CodeQuest — *Participant, NUST*

Odoo × NUST — 8-Hour Technical Workshop Event Details *Oct 2026*

ENGINEERING EXPERIENCE

Viro.pk — *Founder & Full-Stack Developer* Live Site | Code *Mar 2025 – Present*
Next.js • React • Supabase • Tailwind CSS • PWA • REST APIs • Git • Vercel

- Founded, built, deployed, and independently maintain a **production e-commerce platform**, owning its architecture, development, and **release management**.
- Implemented product discovery, **inventory and order management**, **authentication**, analytics, wishlist, WhatsApp integration, cash-on-delivery checkout, and a responsive **PWA**.
- Iterated through **40+ releases**, covering debugging, optimization, and deployment.

MalekPOS — *Client Software Developer* Product Website *Jul – Sep 2026*
Java • JavaFX • Firebase • Client–Server Architecture • Windows (.exe) Packaging

- Developed a **JavaFX** point-of-sale desktop application for a client. It works **online and offline**, supports several terminals through a **client–server** setup, and is packaged as a **Windows executable (.exe)** for installation and distribution.
- Built a supporting website for software downloads and license purchases, with **license management**, terminal upgrades, **expiry-date checks**, and automated billing.
- Integrated **Firebase** for backend data management and automated email notifications.

SELECTED ENGINEERING PROJECTS

OS Kernel Simulator — *Operating Systems Course Project* Demo | Code *Feb – Apr 2026*
Node.js • Express.js • React • MongoDB • JWT • REST APIs

- Built a full-stack OS simulator implementing **CPU scheduling** (FCFS, SJF, Round Robin, Priority) and **page-replacement** algorithms (FIFO, LRU, Optimal), with process synchronization concepts.
- Designed an MVC-oriented **REST** backend with **JWT authentication**, **input validation**, pagination, and centralized error handling.
- Developed interactive **Gantt-chart** visualizations and CSV-based simulation input; deployed frontend and backend via Vercel and Railway.

TECHNICAL SKILLS

Programming Languages: Python • C++ • Java • JavaScript • TypeScript • MATLAB • SQL

LLM Applications & Agents: LangChain (chains, runnables, structured output) • **RAG** (loaders, splitters, embeddings, vector stores, retrievers) • **Tool Calling** • **AI Agents** • LangGraph • **LLM Evaluation** • Ollama

Machine Learning & Statistical Evaluation: Scikit-learn • Data Preprocessing • **Leakage-aware Validation** • Model Comparison • ROC-AUC • PR-AUC • Precision/Recall/F1 • Threshold Optimization • **SHAP** (Explainability)

Deep Learning & Computer Vision: PyTorch • OpenCV • RF-DETR • Object Detection • Image Preprocessing • Data Augmentation • MLPs

ML Deployment & Data Tools: NumPy • Pandas • ONNX Runtime • TensorRT FP16 • FastAPI • Pydantic

Studied Concepts: CNNs • Transfer Learning • RNNs/LSTMs/GRUs • Attention & Transformers • TensorFlow/Keras

Full-Stack & Application Development: React • Next.js • Node.js • Express.js • REST APIs • JavaFX • JWT • Tailwind CSS • PWA

Databases, Tools & Infrastructure: MongoDB • PostgreSQL • MySQL • Supabase • Firebase • **Git** • GitHub • Linux • Postman • Vercel • Railway

RELEVANT COURSEWORK

Mathematics & Scientific Computing: Linear Algebra & ODEs • Probability & Statistics • Calculus & Analytical Geometry • Discrete Mathematics • Numerical Methods • Complex Variables & Transforms

Computer Science & Engineering: Data Structures & Algorithms • Design & Analysis of Algorithms • Operating Systems • Database Systems • Computer Networks • Object-Oriented Programming • Computer Architecture & Logic Design • **Software Engineering** • **Software Design & Architecture**